

EART11200 - The Natural Scientist's Toolkit
Mathematics and Physics

Week 8 Lecture - Integration By Substitution

Last Week...

- We can reverse some common derivatives, we arrive at the following functions.
- The more you practice these procedures, the easier it will become to apply them without looking them up.
- Use your math's survival pack when needed!

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
$cos(x)$	$sin(x) + d$
$sin(x)$	$-cos(x) + d$

Last Week...

- In this course, we focus on Newton's Second Law, which is given by the following equation:

$$F = ma$$

  where F is the net force acting on a body that has a mass m and therefore acceleration a.
- This states acceleration of an object depends on the mass of the object and the amount of force applied.
- Newton's second law can therefore be expressed as an ordinary differential equation:

$$F = m\frac{dv}{dt} \qquad F = m\frac{d}{dt}\left(\frac{dx}{dt}\right) \qquad F = m\frac{d^2x}{dt^2} \qquad F = mv\frac{dv}{dx}$$

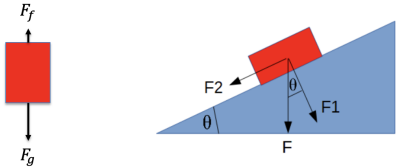
Last Week...

- By examining the forces acting on objects we can formulate ODEs:

$$F = ma \quad \Longrightarrow \quad m\frac{dv}{dt} = mg$$

$$F = ma \quad \Longrightarrow \quad m\frac{dv}{dt} = mg\sin(\theta)$$

- Using separation of variables, we integrate these ODEs to calculate behaviours of our systems.
- Frictional forces will depend on the object, the system, its mass and how quickly it is moving.



Course layout

Date	Week	Topic
27/01/2025	20	Calculating Gradients
03/02/2025	21	Introduction to Differentiation
10/02/2025	22	Differentiating Combinations of Functions
17/02/2025	23	Optimisation & Properties of Derivatives
24/02/2025	24	Second Order Derivatives & Root Finding
03/03/2025	25	Introduction to Integration
10/03/2025	26	Integration using Newtons Laws
17/03/2025	27	Integration by Substitution
24/03/2025	28	Integrating Second Order ODEs
31/03/2025	29	Vectors and Partial Differentiation
07/04/2025	30	Non-teaching week
14/04/2025	31	Easter break
21/04/2025	32	Easter break
28/04/2025	33	Revision: Differentiation
05/05/2025	34	Revision: Integration
Exam Period		

Intended Learning Outcomes

(1) Explain the concept of integration by substitution and its relationship to the chain rule in differentiation.

(2) Apply integration by substitution on indefinite integrals to solve theoretical and real-world problems.

(3) Apply integration by substitution on definite integrals to solve theoretical and real-world problems.

Integration: Common Laws

- In Week 6, we introduced rules that allow us to directly integrate certain basic functions if we can recognize their forms.
- There may be a need to rearrange the equation, often by using techniques like separation of variables (e.g., rearranging N and t).
- Like differentiation, the integral of a sum is equal to the sum of the integrals of the individual functions.

$$\frac{dy}{dx} = u(x) + v(x) \implies y = \int u(x)dx + \int v(x)dx$$

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
ke^{kx}	$e^{kx} + d$
$\frac{1}{x}$	$ln(x) + d$
$cos(x)$	$sin(x) + d$
$sin(x)$	$-cos(x) + d$

Integration: Common Laws

- In some cases, we cant directly integrate by using these rules, as we are faced with more complicated combinations of functions such as **functions of functions**.
- For example, what happens if we want to integrate

$$y = \int \frac{dy}{dx} = \int (3x + 2)^4 dx$$

- This is where **integration by substitution** comes in.
- Integration by substitution is essentially the reverse process of the chain rule in differentiation.

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
constant	$constant \times x + d$
x	$\frac{x^2}{2} + d$
x^n	$\frac{x^{n+1}}{n+1} + d$
e^x	$e^x + d$
ke^{kx}	$e^{kx} + d$
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The Chain Rule

- Use the **Chain Rule** to differentiate:

$$f(x) = \sin(3x + 1)$$

- **Step 1**: Split the function and substitute

$$f(g) = \sin(g) \qquad g(x) = 3x + 1$$

- **Step 2**: Differentiate:

$$\frac{df}{dg} = \cos(g) \qquad \frac{dg}{dx} = 3$$

- **Step 3**: Apply the product rule and substitute back in:

$$f(x) = 3\cos(3x + 1)$$

The Chain Rule

$$\frac{df}{dx} = \frac{df}{dg} \times \frac{dg}{dx}$$

Function $y = f(x)$	Derivative $\frac{dy}{dx}$
constant	0
x	1
x^n	nx^{n-1}
e^x	e^x
e^{kx}	ke^{kx}
$e^{f(x)}$	$\frac{df}{dx}e^{f(x)}$
$ln(x)$	$\frac{1}{x}$
$sin(x)$	$cos(x)$
$cos(x)$	$-sin(x)$

- Let's return to the example where we want to evaluate

$$y = \int (3x + 2)^4 dx$$

- At first glance, it may appear that this resembles the derivative of some function, allowing us to integrate directly.
- What if we replace the expression in the brackets with another variable such that we can integrate using the rules we already know?

$$y = \int u^4 dx$$

Integration by Substitution

- Well, if we are integrating with respect to a new variable u, we need to get rid of all instances of x.
- Since dx relates to x, we need to rewrite it in terms of u.
- This step ensures that the integral is entirely in terms of the new variable.
- When the integral contains a factor like $ax + b$, substituting $u = ax + b$ often simplifies the problem.
- Let's show how integration by substitution works in full.

Integration by Substitution

1. Write out the integral in its simplest form:

$$y = \int (3x + 2)^4 dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = 3x + 2$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = \frac{1}{3}(u - 2)$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = \frac{1}{3}$$

5. Get an expression for dx:

$$dx = \frac{du}{3}$$

6. Make the substitution in the original integral

$$y = \int (3x+2)^4 dx \implies y = \int u^4 \frac{du}{3} \implies y = \frac{1}{3}\int u^4 \, du$$

7. Integrate as usual:

$$y = \frac{1}{3}\int u^4 \, du \implies \frac{1}{15}u^5 + d$$

8. Substitute the original expression back in to get the final expression:

$$y = \frac{1}{15}(3x+2)^5 + d$$

Example: Integration by Substitution

- Lets evaluate the integral

$$y = \int \cos(3x + 4)\, dx$$

- Let's use $u = 3x + 4$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = 3x + 4 \implies x = \frac{1}{3}(u - 4) \implies \frac{dx}{du} = \frac{1}{3} \implies dx = \frac{1}{3}du$$

- Use these to make the substitution:

$$\int \cos(3x + 4)\, dx \implies \frac{1}{3}\int \cos(u)\, du$$

Example: Integration by Substitution

- Now we can evaluate using the basic laws of integration:

$$\frac{1}{3}\int \cos(u)\,du = \frac{1}{3}\sin(u) + c$$

- Substituting $u = 3x + 4$ back in gives us:

$$y = \frac{1}{3}\sin(3x + 4) + c$$

- You can differentiate using the chain rule to verify that this works.

Derivative $\frac{dy}{dx}$	**Integral** y
0	d
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Integration by Substitution

- This is integration by substitution for **indefinite integrals**.
- In many cases, we want to find the **area under a curve** so the integral has limits:

$$y = \int_{-1}^{1} (3x + 2)^4 dx$$

- We need to perform **definite integration** in this case.
- We calculate the integral (using integration by substitution) then evaluate the integral using new limits under the substitution.
- Let's show how definite integration by substitution works in full.

Integration by Substitution

1. Write out the integral in its simplest form:

$$y = \int_{-1}^{1} (3x + 2)^4 dx$$

2. Find a new variable (u) as a function of the old variable (x) as a substitution:

$$u = 3x + 2$$

3. Rearrange to get an expression for the old variable (x) in terms of the new variable (u):

$$x = \frac{1}{3}(u - 2)$$

4. Differentiate with respect to the new variable u:

$$\frac{dx}{du} = \frac{1}{3}$$

5. Get an expression for dx:

$$dx = \frac{du}{3}$$

6. Find the new limits of the integral:

$$\text{When } x = -1, \quad u = 3 \times (-1) + 2 = -1$$
$$\text{When } x = 1, \quad u = 3 \times 1 + 2 = 5$$

7. Make the substitution in the original integral

$$y = \int_{-1}^{1} (3x+2)^4 dx \implies y = \int_{-1}^{5} u^4 \frac{du}{3} \implies y = \frac{1}{3}\int_{-1}^{5} u^4 \, du$$

8. Integrate as usual:

$$\frac{1}{3}\int_{-1}^{5} u^4 \, du = \frac{1}{3}\left[\frac{1}{5}u^5\right]_{-1}^{5} = \frac{1}{3} \times \frac{1}{5} \times (5^5 - 1^5) = \frac{3124}{15}$$

Example: Integration by Substitution

- Lets evaluate the integral

$$y = \int_{-9}^{1} \frac{1}{x+10} dx$$

- Let's use $u = x + 10$ as a substitution. Lets rearrange for x, differentiate in terms of u and find an expression for dx:

$$u = x + 10 \implies x = u - 10 \implies \frac{dx}{du} = 1 \implies dx = du$$

- Find the new limits:

$$\text{When } x = -9, \quad u = -9 + 10 = 1$$
$$\text{When } x = 1, \quad u = 1 + 10 = 11$$

Example: Integration by Substitution

- Use these to make the substitution:

$$\int_{-9}^{1} \frac{1}{x+10} dx \implies \int_{1}^{11} \frac{1}{u} du$$

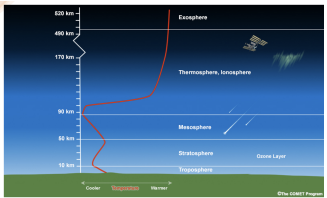
- Now we can evaluate using the basic laws of integration:

$$\begin{aligned} \int_{1}^{11} \frac{1}{u} du &= [\ln(u)]_{1}^{11} \\ &= \ln(11) - \ln(1) \\ &= \ln(11) = 2.398 \end{aligned}$$

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Example: Integration by Substitution

- So we see that we can integrate more complicated functions using substitution.
- We substitute so that the integration becomes simple so we can use our table of integrals.
- Why does this matter? If the physics of the system requires another formulation, a substitution might be needed.
- For example, the Hydrostatic Equation, but with Temperature varying with altitude:



$$\frac{dP}{dz} = -\frac{P}{R_a\left(T_{surface} - 0.01z\right)}g \quad \Longrightarrow \quad \int \frac{1}{P}dP = -\frac{g}{R_a}\int \frac{1}{T_{surface} - 0.01z}dz$$